From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

FIITJEE

JEE(Main)-2014

ALL INDIA INTEGRATED TEST

ANSWERS, HINTS & SOLUTIONS

PART TEST–III

(Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	D	A	C
2.	C	A	A
3.	C	B	C
4.	A	A	C
5.	D	C	D
6.	C	D	B
7.	B	C	A
8.	B	C	A
9.	C	C	C
10.	A	A	B
11.	D	A	D
12.	C	A	B
13.	C	B	D
14.	A	A	A
15.	A	A	B
16.	A	C	A
17.	D	A	B
18.	B	B	B
19.	B	B	C
20.	B	A	C
21.	B	C	C
22.	A	C	B
23.	D	C	B
24.	B	B	B
25.	A	C	C
26.	A	B	D
27.	D	A	B
28.	D	B	D
29.	A	B	A
30.	A	B	B

Physics

PART – I

SECTION – A

1. Field will be in opposite direction in the two halves and infinite at x close to zero & L.

2. Calculate net resistance, power = E^2/R_{net}

5. Force due to $\vec{B}$ should be in $-\hat{j}$ direction and due to $\vec{E}$ should be in $\hat{i}$ direction.

6. Use Ampere's circuital law

7. Use Gauss Law

8. Calculate equivalent internal resistance by using series and parallel grouping of cells.

9. Solve as balanced Wheatstone bridge and resistance of wire will be proportional to lengths.

14. $\vec{\mu} = \left(\frac{\pi R^2}{2} I\right)\hat{i} + \left(\frac{\pi R^2}{2} I\right)\hat{j} = \frac{\pi R^2 I}{2}(\hat{i} + \hat{j})$

15. $v_x = v$ = constant

$q\int E\cos\theta dt = mv_y$

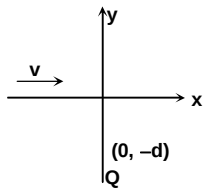
or $q\int E\cos\theta \frac{dA}{2\pi dv} = mv_y$

$v_y = \frac{q}{2\pi mdv}\int E\cos\theta dA$

from Gauss' law

$\int E\cos\theta dA = \frac{q}{\varepsilon_0}$ Integration for the whole cylindrical surface of radius d and extending from $x \to -\infty$ to $x \to +\infty$

Deflection $\theta = \tan^{-1}\left(\frac{qQ}{2\pi\varepsilon_0 dmv^2}\right)$



16. $\vec{E} = -\frac{kqd}{y^3}\hat{i} - \frac{2kqd}{y^3}\hat{j}$

29. $Q = 4\pi\varepsilon_0 RV$

$4\pi R^2 t = \frac{4}{3}\pi r^3$

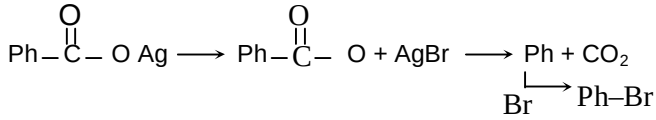
$V_{required} = \frac{Q}{4\pi\varepsilon_0 r}$ = 3 volts.

Chemistry

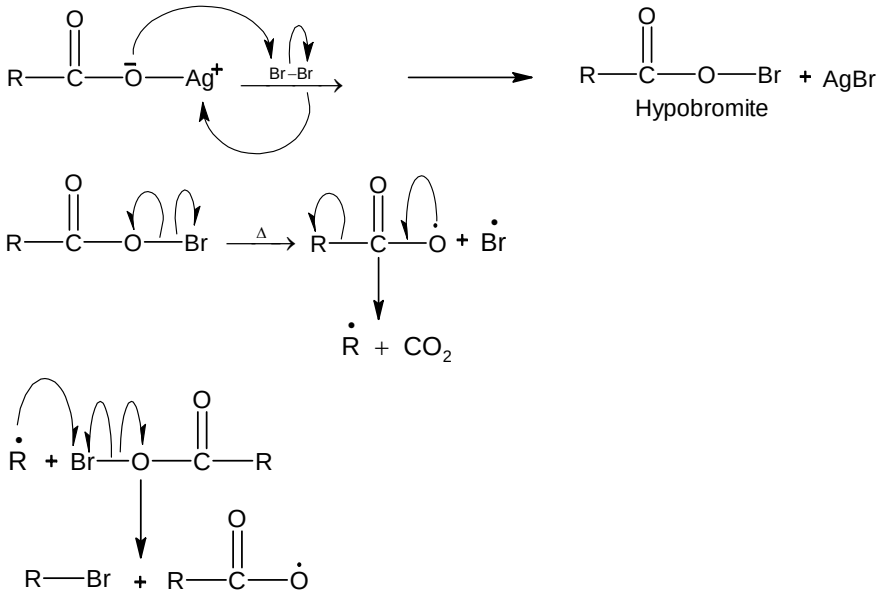
PART – II

SECTION – A

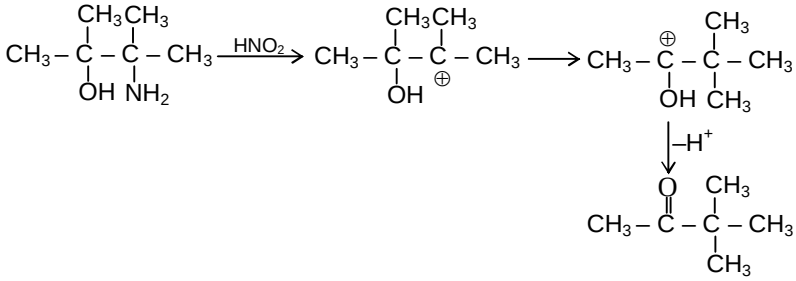
1. This is Hundsdicker reaction.



2.

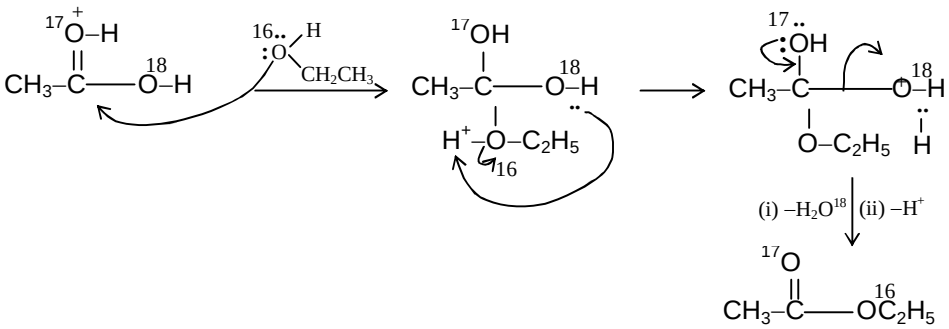


3.

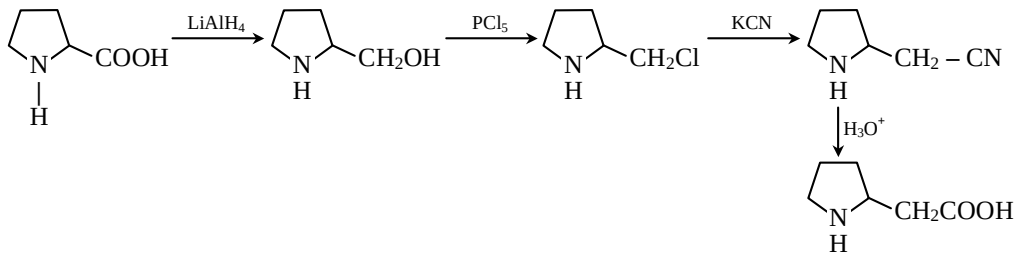


4. Carbylamine reaction is given by primary amines only.

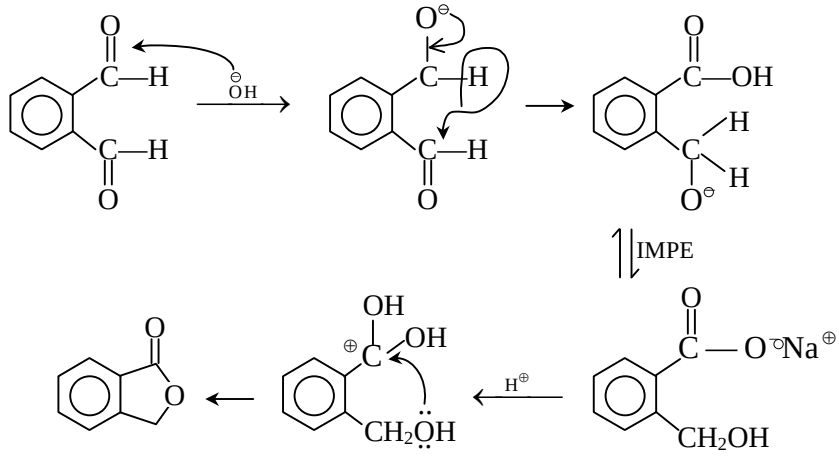
5.



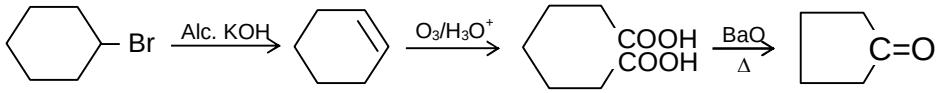
6.



7.



8.

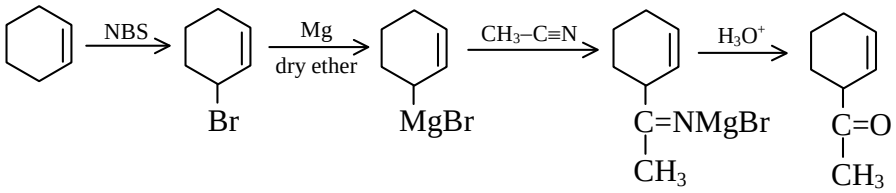


9. Test of terminal alkynes.

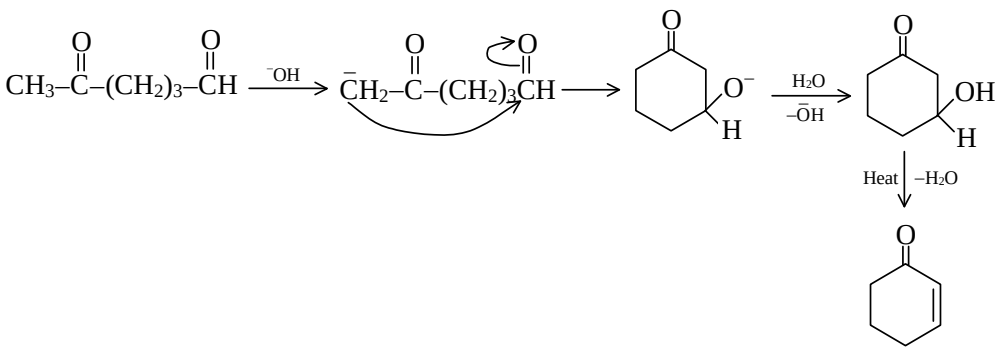
11. In polystyrene the monomeric unit is styrene which is having the formula of $H_2C{=}CH{-}C_6H_5$.

12. $n\ CH_2{=}CH{-}CH{=}CH_2 + n\ C_6H_5{-}CH{=}CH_2$ (styrene) $\rightarrow [CH_2{-}CH{=}CH{-}CH_2{-}CH(C_6H_5){-}CH_2]_n$
(1,3-butadiene) (Buna–S)

13.

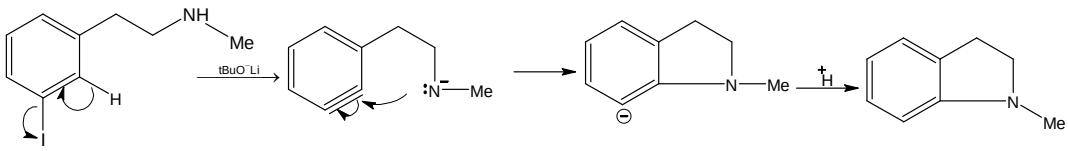


14.



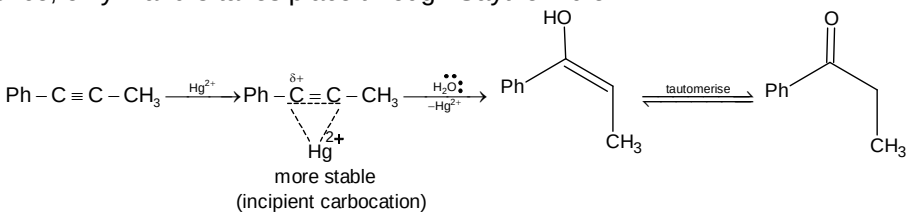
15. Due to the presence of CS_2, only monobromination takes place as CS_2 decrease the electrophilicity of Br_2 and also minimize phenol ionization.

18.

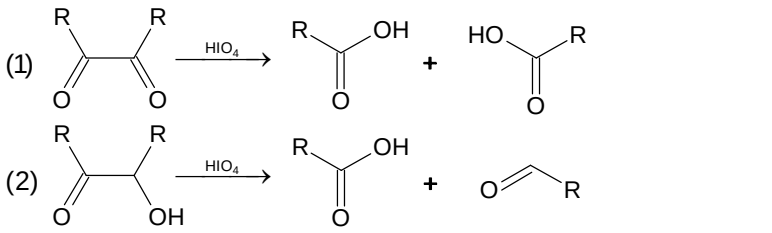


19. (1) proceeds through E2 path while (2) and (3) proceeds through E1 path and (4) proceeds through E1CB path.
Hence, only 2 and 3 takes place through Saytzeff rule.

20.

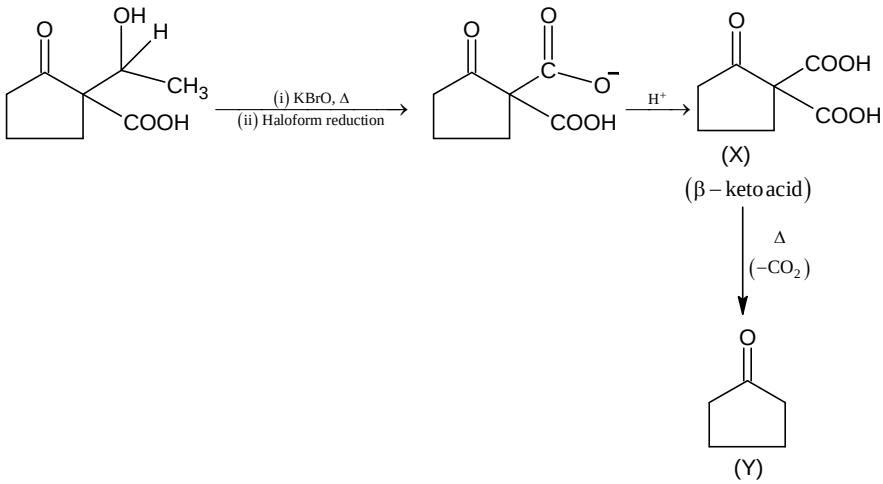


22.

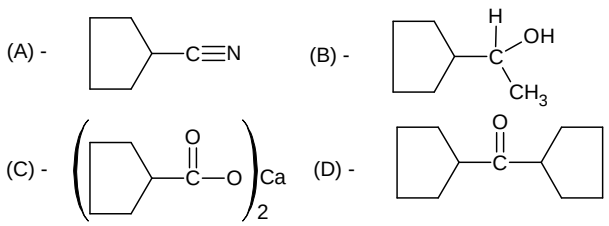


(4) $R-CHOH-CHOH-R \xrightarrow{HIO_4} R-CH=O + R-CH=O$

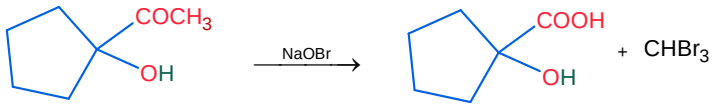
23.



25.



26. This is an example of Cross – Cannizzaro reaction. The molecule first attacked is converted to sodium salt of benzoic acid in a Cannizzaro reaction. The presence of $-NO_2$ group activates the benzene ring towards nucleophilic attack. Therefore nitrobenzaldehyde is preferentially attacked.

30. 

Haloform reaction

Mathematics PART – III

SECTION – A

1. $f(2x) = f(x) = f(x/2) = f(x/4) = \ldots = f\left(\frac{x}{2^{n-1}}\right) = f\left(\frac{x}{2^n}\right)$

consider $f(x) = f\left(\frac{x}{2^n}\right)$

$\lim_{n\to\infty} f(x) = \lim_{n\to\infty} f\left(\frac{x}{2^n}\right)$

$\Rightarrow f(x) = f(0)\ \forall\ x \in R$

(since f(x) is given to be a continuous function)

$\Rightarrow f(x) = f(2010) = \frac{\pi}{2}$

$\Rightarrow \lim_{x\to 0} \frac{\cos\frac{\pi}{2} - \sqrt[3]{\cos\frac{\pi}{2}}}{\sin^2 x} = 0$

2. $\frac{1}{(1+x)^2} \le \frac{1}{\sqrt{1+x^3}} \le \frac{1}{1+x}$ for $x \in [2, 3]$

$\Rightarrow \int_2^3 \frac{1}{(1+x)^2}dx \le \int_2^3 \frac{dx}{\sqrt{1+x^3}} \le \int_2^3 \frac{dx}{1+x}$

$\frac{1}{12} \le \int_3^2 \frac{dx}{\sqrt{1+x^3}} \le \ln\frac{4}{3} < 1$

$\Rightarrow \left|\left[\int_2^3 \frac{dx}{\sqrt{1+x^3}}\right]\right| = 0$

3. $\int \frac{x\tan^{-1}x}{(1+x^2)^{3/2}}dx$. Let $\tan^{-1}x = t \Rightarrow \frac{dx}{1+x^2} = dt$

$= \frac{x - \tan^{-1}x}{\sqrt{1+x^2}} + c$ now compare it with RHS $k_1 = -1$, $k_2 = -x$ hence $\int_0^2 \max\{-1, -x\}dx = \frac{-3}{2}$

4. $(f(x))^3 + (x)^3 + (-1)^3 - 3(-1)(x)(f(x)) = 0$

$\Rightarrow (f(x) + x + (-1))\left(f^2(x) + x^2 + (-1)^2 - xf(x) + x + f(x)\right) = 0$

Using $a^3 + b^3 + c^3 - 3abc = (a+b+c)\left(a^2+b^2+c^2-ab-bc-ca\right)$

$f^2(x) + x^2 + (-a)^2 - xf(x) + x + f(x)$

$= \frac{1}{2}\left\{(f(x)-x)^2 + (x+1)^2 + (f(x)+1)^2\right\} \ge 0$

Equality occurs at $f(x) = x = -1 \Rightarrow f(1) = -1$

But it is given that $f(1) \ne -1$

Only solution is $f(x) = 1 - x$

domain of $\sqrt{1-f(x)} = \sqrt{x}$ is $x \in R^+ \cup \{0\}$

5. when $x \to a^-$
$[x] - x \to -1$
and $f(x)\big|_{\text{at } x=a^-} = x - a, f'(x)\big|_{\text{at } x=a^-} = 1$

6. $\int_0^x f^{-1}(x)\,dx = \int_{f^{-1}(0)}^{f^{-1}(x)} tf'(t)\,dx = tf(t)\big|_{f^{-1}(0)}^{f^{-1}(x)} - \int_{f^{-1}(0)}^{f^{-1}(x)} f(t)\,dt$ (by substituting x = f (t))

$\Rightarrow \int_0^{\pi} f^{-1}(x)\,dx = \pi^2 - \int_0^{\pi}(x + \sin x)\,dx = \frac{\pi^2}{2} - 2$

7. Since f (x) and f′ (x) are symmetric about the line y = – x.
If (α, β) lies on y = f (x) then $(-\beta, -\alpha)$ on $y = f^{-1}(x)$
$\Rightarrow (-\alpha, -\beta)$ lies on y = f (x)
$\Rightarrow$ y = f (x) is odd.

8. f(3) = 3f(1) = 3, f(4) = f(2 + 1 + 1) = 2 + 1 + 1 = 4 and so one.
In general, we get f(r) = r for $r \in N$

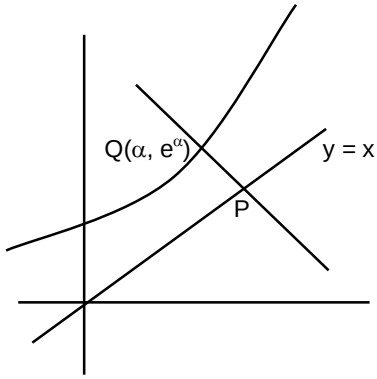
$\Rightarrow \lim_{n\to\infty} \frac{\sum_{r=1}^{n}(4r)f(3r)}{n^3} = \lim_{n\to\infty} \frac{12n(n+1)(2n+1)}{6n^3} = 4$

9. $\int e^{g(x)}\left[g'(x)f(x) + f'(x)\right]dx$

10. Clearly α is repeated root of f(x) = 0
$\lim_{x\to\alpha} \frac{f(x)}{f'(x)}\left(\frac{f'(x)}{f(x)} - \left\{\frac{f'(x)}{f(x)}\right\}\right) \Rightarrow \lim_{x\to\alpha}\left(1 - \frac{f(x)}{f'(x)}\left\{\frac{f'(x)}{f(x)}\right\}\right)$ = 1–0=1

$\left(\lim_{x\to\alpha} \frac{f(x)}{f'(x)} = 0 \;\&\; \left\{\frac{f'(x)}{f(x)}\right\} \text{is bounded function}\right)$

11. We know the shortest and greatest distance between two curves always lie along the common normal. Let the normal to the $y = e^x$ at $Q(\alpha, e^{\alpha})$ meets the line y = x at P.



$\frac{dy}{dx} = e^x$

$\therefore -1 = \left(-\frac{1}{\frac{dy}{dx}}\right)_{x=\alpha} \Rightarrow -e^{-\alpha} = -1$

$\Rightarrow \alpha = 0$
$\therefore$ Q ≡ (0, 1)
$\therefore$ normal = y – 1 = – 1(x – 0) $\Rightarrow$ x + y – 1 = 0
$\therefore$ $P = \left(\frac{1}{2}, \frac{1}{2}\right)$
$\therefore$ $PQ = \sqrt{\frac{1}{4} + \frac{1}{4}} \Rightarrow \frac{1}{\sqrt{2}}$

12. $y^2 - xy = \left(y - \frac{x}{2}\right)^2 - \frac{x^2}{4} \Rightarrow y^2 - xy$ is decreasing to $y \le \frac{x}{2}$ and increasing to $y \ge \frac{x}{2}$.

Thus the largest value of $y^2 - xy$ must be at $y = 0, \frac{x}{2}$ or 1

The values are $0, \frac{x^2}{4}, |1 - x| \Rightarrow$ for $x \notin (0, 1)$ $g(x) \geq \frac{1}{4}$

and for $x \in (0, 1)$ $g(x) = \max\left(\frac{x^2}{4}, 1-x\right)$

also, $x^2 + 4x - 4 = 0 \Rightarrow x = -2 \pm \sqrt{8} \Rightarrow g(x) = 1 - x$ for $x \leq \sqrt{8} - 2$ and $\frac{x^2}{4}$ for $x \geq \sqrt{8} - 2$

$\Rightarrow g(x)$ is minimum at $x = \sqrt{8} - 2$ and minimum value is $3 - \sqrt{8}$

13. $f(x) = |\sin x| + |\cos x|$ has period $\frac{\pi}{2}$

$g(x) = \sin ax + \cos ax$

$g(x) = \sqrt{2}\sin\left(ax + \frac{\pi}{4}\right)$

g(x) has period $\frac{2\pi}{|a|} = \frac{\pi}{2}$

$|a| = 4 \quad a = 4$

14. $P(x) - 5 = (x - a)(x - b)(x - c)(x - d)q(x)$
$\Rightarrow |P(x) - 5| = |x - a||x - b||x - c||x - d||qx|$
$\geq |(1)(-1)(2)(-2)(1)| \geq 4$
$\Rightarrow P(x) \geq 5 + 4$ or $\leq 5 - 4 \Rightarrow P(x) \geq 9$ or ≤ 1.

15. Dividing the numerator and denominator by x^2, the given integral becomes

$$\int \frac{\left(1-\left(\frac{1}{x^2}\right)\right)dx}{\left[\left(x+\left(\frac{1}{x}\right)\right)^2+1\right]\tan^{-1}\left(x+\left(\frac{1}{x}\right)\right)}$$

Let $x + \frac{1}{x} = \tan v = \int \frac{dv}{v} = \log|v| + c$

$= \log\left|\tan^{-1}\left(\frac{x^2+1}{x}\right)\right| + c$. Hence $k = 1$

16. $f(x) = \frac{\pi}{2} + \sin^{-1}x + 3\tan^{-1}x$

As $\sin^{-1}x + 3\tan^{-1}x$ is increasing in its domain $x \in [-1, 1]$
so, $[f(-1), f(1)]$

$f(-1) = \frac{\pi}{2} - \frac{\pi}{2} - \frac{3\pi}{4} = -\frac{3\pi}{4}$

$f(1) = \frac{\pi}{2} + \frac{\pi}{2} + \frac{3\pi}{4} = \frac{7\pi}{4}$.

17. As clear from the graph of $\tan^{-1}|x|$ and $-\cos^{-1}x$, there is only one solution.

18. $I = \int \frac{x^2\left(1-\frac{1}{x^2}\right)dx}{x^2\left(x+\frac{1}{x}\right)\left(x^2+\frac{1}{x^2}\right)^{1/2}}$

Let $x+\frac{1}{x}=p \Rightarrow \left(1-\frac{1}{x^2}\right)dx = dp$

$I = \int \frac{dp}{p\sqrt{p^2-2}} = \frac{1}{\sqrt{2}}\sec^{-1}\frac{p}{\sqrt{2}} = \frac{1}{\sqrt{2}}\sec^{-1}\left(\frac{x^2+1}{\sqrt{2}x}\right)+c$

19. $\int_0^{\infty}\left[\frac{3}{x^2+1}\right]dx = \int_0^{\frac{1}{\sqrt{2}}} 2dx + \int_{\frac{1}{\sqrt{2}}}^{\sqrt{2}} 1\cdot dx + \int_{\sqrt{2}}^{\infty} 0\cdot dx = \frac{3}{\sqrt{2}}$

20. $3-2^x-2^{1-x} \geq 0$ & $\sin^{-1}x \geq 0$

$\Rightarrow 1 \leq 2^x \leq 2$ & $x \in [0, 1]$

$\Rightarrow 0 \leq x \leq 1$ & $x \in [0, 1]$

$\Rightarrow x \in [0, 1]$

21. Let $f^{-1}(x) = g(x)$

$f(g(x) = x$

$g'(x) = \frac{1}{f'(g(x))} = \frac{1}{f'(g(3))} = \frac{1}{f'(0)} = \frac{1}{\sin^2(\sin 1)}$.

22. $f(x) = Ax^3 + Bx^2 + Cx + D$

$f'(x) = 3Ax^2 + 2Bx + C$

$f''(x) = 6Ax + 2B$

$\Rightarrow f''(x)$ changes its sign only at one point i.e. $x = -B/3A$

23. Put $\sin^2 x + \cos^3 x = t$.

24. $\lim_{x\to 0} \frac{(1-\cos x^6)(1+\cos x^6+\cos^2 x^6+.....\cos^{11} x^6)}{\frac{\sin x \sin 2x \sin 3x....\sin 12x}{(x.2x.3x....12x)}\cdot(x.2x.3x....12x)}$

$= \frac{1}{2\times(12)!}\times 12 = \frac{1}{2(11)!}$.

25. $\lim_{x\to 0}\left\{\frac{\sin x}{x}\right\}^{\frac{1}{\left\{\frac{\tan x}{x}\right\}}} = \lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{\frac{\tan x}{x}-1}} = e^{-\frac{1}{2}}$

26. Let $g(x) = ax^3+bx^2 + cx + d$

$g'(x) = 3ax^2+2bx+c$

have zeroes $x = 2, -2$

Now a is a root of $x^2-x-6=0$

$\Rightarrow a = 3$ or -2

When $a = 3$, $c = -36$ and $a = -2$, $c = 24$

Now $g(-2) > 0$
$-9a - 2c + d > 0$
$D > 32$

27. $f'(b) = 0$ and $f(b + x) = f(b - x)$

28. $I = \int_0^1 e^{x^{2008}} dx = x.e^{x^{2008}}\Big|_0^1 - \int_0^1 2008x^{2008}.e^{x^{2008}} dx$

$\Rightarrow \int_0^1 \left(1+\left(2008\right)x^{2008}\right)e^{x^{2008}} dx = e$

29. $f(x) = e^{-x^2}$

$\ell n f(x) = -x^2$

$-\ell n f(x) = x^2$

$\sqrt{-\ell n f(x)} = x$

$\int_0^1 f(x)dx + \int_1^{1/e} f^{-1}(x)dx$

$= 1\times\frac{1}{e} - 0\times 1 = \frac{1}{e}.$

30. $y = \underset{n\to\infty}{Lt}\left\{\left(\frac{1}{n}\right)\left(\frac{2}{n}\right)^2\left(\frac{3}{n}\right)^3 .....\left(\frac{n}{n}\right)^n\right\}^{\frac{1}{n^2}}$

$\log y = \frac{\sum_{r=1}^{n} r\log(r/n)}{n^2}$

$= \int_0^1 x \log x \, dx$

$y = e^{-\frac{1}{4}}$